Mike Hall

Staff Software Engineer / Systems Architect

Chicago, IL | mike@just3ws.com | (847) 877-3825 | just3ws.com | linkedin.com/in/just3ws/ | github.com/just3ws

Professional Summary

Staff Software Engineer specializing in high-consequence legacy modernization, distributed systems architecture, and platform resilience. Combines hands-on software engineering with platform architecture, OpenTelemetry distributed tracing, cross-lane boundary mediation, and zero-downtime data migrations across Ruby on Rails, PostgreSQL, TypeScript, and cloud infrastructure.

Core Skills

Platform Architecture & Legacy Modernization, Cross-Domain Boundary Architecture & Governance, OpenTelemetry & Distributed Trace Context Propagation, High-Consequence Incident Escalation & Root Cause Analysis, Zero-Downtime Data Migrations & Schema Partitioning, Ruby on Rails, PostgreSQL, pgvector, Sidekiq, Solid Queue & Redis, TypeScript, Node.js, C#, .NET, Java & SQL, AWS, Docker, Kubernetes & CI/CD Pipelines, Privacy-Conscious Local AI Orchestration & RAG Tooling

Experience

Staff Software Engineer

just3ws (Independent) | Chicago, IL | February 2026 - Present

Architected and operating sovereign production platforms across Ruby on Rails 8, PostgreSQL 18 (pgvector), Solid Queue, TypeScript, and local Metal GPU AI inference runtimes, combining autonomous media pipelines, deterministic 5-stage FFmpeg audio DSP restoration (dehum), hybrid vector/lexical search, deterministic state engines (Phalanx Duel), and multi-source ingestion with prompt-injection guardrails (WWWorkRemote).

- **Autonomous Media Pipeline & Local AI RAG Microservice:** Architected a Rails 8, Solid Queue, and PostgreSQL 18 (pgvector) media ingestion and publishing platform with on-device Metal GPU Whisper transcription, hybrid lexical and vector semantic search via Reciprocal Rank Fusion (RRF), local LLaMA 3.1 RAG synthesis via RubyLLM, OpenTelemetry distributed tracing, Ahoy analytics, and PgHero query tuning across 270+ processed video masters.
- **Forensic Audio DSP Restoration & Sovereign Verification (dehum):** Engineered a deterministic 5-stage FFmpeg audio restoration and clock-drift synchronization pipeline (atempo, adeclick, highpass, afftdn, and 8-harmonic 60 Hz notch cascade) eliminating electromagnetic interference and corona discharge crackle recorded under 345,000-volt transmission lines while preserving lossless video streams.
- **Technical Conversation Archive & Knowledge Graph:** Built and curated a contract-validated oral history platform preserving 218 technical interviews with Software Craftsmanship, Ruby, and JVM practitioners, enforcing build-time schema verification, speaker diarization, and semantic knowledge graph cross-linking across 500+ indexed pages.
- **Deterministic State Engine & Verification Platform (Phalanx Duel):** Engineered a server-authoritative real-time tactical game platform in TypeScript, Node.js, PostgreSQL, and WebSockets featuring a pure deterministic rules engine, append-only replayable action ledger, OpenTelemetry trace propagation, and property-based CI verification gates across the complete state lifecycle.
- **Multi-Source Ingestion & Prompt-Injection Defense (WWWorkRemote):** Designed a Rails 8 and pgvector semantic matching platform with a Manifest V3 Chrome extension across 16 ATS adapters and a 4-stage prompt-injection guardrails pipeline screening untrusted scraped text before local LLM evaluation.

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Led modernization of the Acquisition Lane, the digital front door for customer application and origination funnel workflows, delivered an instant disbursement workflow, applied Architecture Discovery to a silent application-completion defect, and aligned distributed observability with SRE operations.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.

- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.
- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Modernized search and asynchronous messaging pipelines under live production load, executing zero-downtime Rails framework upgrades and database relational indexing.

- **Search Infrastructure Modernization:** Eliminated external cluster failure modes and reduced operating footprint by migrating legacy Sphinx to optimized MySQL full-text search and relational indices under live production traffic.
- **Zero-Downtime Rails Upgrades:** Led Ruby and Rails framework upgrades across core applications, sequencing database migrations to ensure zero downtime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Improved testability and performance inside a tightly coupled PHP/MySQL CRM backend, introducing structured query patterns and cache strategies, and hardening the Ember.js frontend test suite across core contact management workflows.

- **Legacy Query Modernization:** Replaced ad-hoc global data access with structured, cacheable query patterns in the PHP backend, enabling unit testability and sub-second query execution.
- **Frontend Verification Gates:** Hardened Ember.js automated test suites across core CRM contact workflows, creating safety gates that enabled non-breaking legacy PHP refactoring.

Senior Software Developer

BenchPrep | Chicago, IL | March 2017 - February 2018

Owned enterprise assessment workflows, optimizing complex PostgreSQL scoring calculations and implementing JWT cryptographic payload verification across embedded client APIs.

- **Assessment Query Optimization:** Optimized complex PostgreSQL calculations across high-stakes exam scoring engines, eliminating calculation drift and reducing query latency.
- **JWT Response Verification:** Implemented JWT-based cryptographic payload signing and once-only processing, securing exam submission pipelines against client-side tampering.
- **Secure Embedded Client APIs:** Designed secure cross-origin API boundaries for embedded enterprise assessment clients, establishing explicit authentication and trust contracts.

Senior Software Developer

ReachLocal | Remote | March 2015 - November 2016

Architected a phased Strangler-Fig modernization roadmap and internal API Gateway with token authentication for a high-volume digital marketing platform.

- **Strangler-Fig Modernization:** Architected incremental modernization roadmap, replacing a high-risk full rewrite with a phased Strangler-Fig migration under production load.
- **Internal API Gateway:** Built internal API Gateway with standardized JSON contracts and token authentication, safely isolating legacy backend services.
- **Automated CI/CD Verification:** Introduced automated static analysis and security scanning into deployment pipelines, preventing regression and vulnerability drift.

Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems)

Groupon | Chicago, IL | July 2011 - May 2013

Implemented a centralized merchant taxonomy service in Java and MySQL, engineered analytical fraud detection in Vertica, and led global technical onboarding curricula during hyper-growth.

- **Centralized Taxonomy Service:** Implemented merchant taxonomy service in Java and MySQL, eliminating categorization drift across distributed product teams during hypergrowth.
- **Analytical Fraud Detection:** Built analytical fraud detection queries in Vertica and conducted exploratory spikes on Hadoop and Clojure for transaction anomaly scoring.
- **Merchant Analytics Pipelines:** Built high-throughput merchant analytics tools in Ruby and CouchDB, surfacing market insights for global sales operations.
- **Global Engineering Enablement:** Redesigned technical onboarding curricula, standardizing engineering practices and shortening time-to-first-commit for 100+ global engineering hires.

Software Engineer

Obtiva | Chicago, IL | August 2009 - July 2011

Delivered backend systems and data-driven architectures as a consultant for Sears, Leapfrog Online, and Groupon during extreme user growth.

- **Self-Service Analytics Platforms:** Built reporting platforms for Leapfrog Online, enabling business stakeholders to query and export data warehouse analytics autonomously.
- **Enterprise B2B Catalog Services:** Developed commercial B2B sales and product catalog integration services for Sears, opening new enterprise revenue streams.
- **Hypergrowth Transaction Pipelines:** Engineered deal-processing services and fraud analysis pipelines at Groupon, stabilizing backend transaction infrastructure during extreme user growth.

Earlier Career Experience

- **Principal Consultant** | **Tandem** | Chicago, IL | August 2018 - August 2018: Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.
- **CTO** | **KloboMedia** | Remote | March 2014 - September 2016: Owned end-to-end platform architecture, leading high-throughput data integration for an early-stage social media startup. The product, TheSocReport, ended when the upstream data taps closed: Twitter's firehose shutdown and the closing of Facebook's and Instagram's user-intelligence APIs removed the inputs the analytics depended on.
- **Open-Source Transition Lead** | **Coderwall** | Remote | January 2014 - December 2014: Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- **Senior Software Developer** | **Upcity** | Remote | October 2013 - February 2014: Owned payment integrations and local development infrastructure, leading system modernization for an SOA-based marketing platform.
- **Senior Software Developer** | **Viewpoints** | Chicago, IL | May 2013 - October 2013: Owned core business system development, leading AWS deployment modernization and CI implementation.
- **Principal Developer** | **Business Decisions, Inc.** | Remote | October 2008 - August 2009: Owned all technical strategy and system development, leading lifecycle modernization for multi-firm consulting teams.
- **Assistant Vice President, Application Developer** | **JPMorgan Chase & Co.** | Chicago, IL | April 2008 - October 2008: Owned cross-system integration stability, leading defect resolution in regulated financial workflows.
- **Senior .NET Developer** | **Brightstar Corporation** | Libertyville, IL | March 2007 - April 2008: Owned enterprise supply-chain integration, leading cross-system data unification for a global mobile device distributor.
- **Senior .NET Developer** | **TicketsNow** | Crystal Lake, IL | November 2005 - March 2007: Owned real-time inventory systems, leading transactional integrity and iterative delivery for revenue-critical operations.
- **Senior Software Developer** | **BP** | Naperville, IL | June 2005 - November 2005: Owned critical feature enhancements for an industrial invoicing system, leading performance optimization in high-volume billing workflows.
- **Senior Software Developer** | **Motorola** | Libertyville, IL | August 2004 - May 2005: Owned retail feature delivery, leading iterative execution in a complex enterprise sales portfolio application.

Technical Leadership

- Established enterprise-wide OpenTelemetry standards and transitioned ownership after adoption stabilized across teams

- Realigned Acquisition engineering responsibilities around business ownership, then founded and technically led an enablement team to sustain platform risk and remediation work
- Built Communities of Practice to prevent architectural drift and align cross-team decisions
- Built and ran global engineering onboarding and training at Groupon during hypergrowth, improving ramp time and retention
- Founded Software Craftsmanship McHenry County (2009) with Ryan Gerry and Jim Suchy, building a community that still meets with 820+ members
- Founded WHOIS Tech Community (UGtastic), interviewing industry leaders and shaping software craftsmanship discourse